

SQL JOIN Practice Tasks

For Beginners - Oracle HR Schema - Tank54 Group

Tables You Need

Table	Key Columns	Description
employees	employee_id, first_name, last_name, salary, department_id, manager_id, job_id, hire_date	All workers in the company
departments	department_id, department_name, manager_id, location_id	Company departments
jobs	job_id, job_title, min_salary, max_salary	Job roles and salary ranges
locations	location_id, street_address, city, country_id	Office locations
countries	country_id, country_name, region_id	Country information
regions	region_id, region_name	World regions
job_history	employee_id, start_date, end_date, job_id, department_id	Past jobs of employees

Part 1: INNER JOIN

INNER JOIN returns only rows that have a match in BOTH tables. If a row has no match, it does NOT appear in the result. This is the most common JOIN type.

Task 1. Basic Two-Table JOIN

Difficulty: Easy

Write a query to show each employee's **first name**, **last name**, and their **department name**. Join the **employees** table with the **departments** table using **department_id**.

Expected columns: first_name, last_name, department_name | Tables: employees, departments

Your Answer:

Task 2. JOIN with ORDER BY

Difficulty: Easy

Write a query to show the **employee last name** and **department name**. Order the result by employee **salary** from highest to lowest (DESC).

```
Expected columns: last_name, department_name | Tables: employees,  
departments | Extra: ORDER BY
```

Your Answer:

Task 3. JOIN with WHERE Clause

Difficulty: Easy

Write a query to show the **employee last name**, **job title**, and **salary**. Join **employees** with **jobs**. Show only employees who earn **more than 12,000**.

```
Expected columns: last_name, job_title, salary | Tables: employees, jobs |  
Extra: WHERE salary > 12000
```

Your Answer:

Task 4. Three-Table INNER JOIN

Difficulty: Medium

Write a query to show the **employee last name**, the **department name**, and the **job title**. You need to join 3 tables: **employees**, **departments**, and **jobs**.

```
Expected columns: last_name, department_name, job_title | Tables:  
employees, departments, jobs
```

Your Answer:

Task 5. Multi-Table JOIN with WHERE

Difficulty: Medium

Write a query to show the **employee last name**, **department name**, and **city** (from the locations table). Show only departments located in the city **Seattle**. You need to join 3 tables: **employees**, **departments**, and **locations**.

```
Expected columns: last_name, department_name, city | Tables: employees, departments, locations | Extra: WHERE city
```

Your Answer:

Task 6. Four-Table JOIN with ORDER BY

Difficulty: Medium

Write a query to show the **employee last name**, **job title**, **department name**, and **city**. Join 4 tables: **employees**, **jobs**, **departments**, and **locations**. Order the result by **salary** from highest to lowest.

```
Expected columns: last_name, job_title, department_name, city | Tables: employees, jobs, departments, locations | Extra: ORDER BY salary DESC
```

Your Answer:

Part 2: LEFT (OUTER) JOIN

LEFT JOIN returns ALL rows from the left table. If there is no match in the right table, the result shows **NULL** for the right table columns. This is useful when you want to keep all records from one side, even without a match on the other side.

Task 7. Basic LEFT JOIN

Difficulty: Easy

Write a query to show the **employee last name** and **department name** for ALL employees. Use LEFT JOIN so that employees without a department also appear (their department_name will be NULL).

```
Expected columns: last_name, department_name | Tables: employees LEFT JOIN departments
```

Your Answer:

Task 8. Find Rows with No Match

Difficulty: Easy

Write a query to find employees who do **NOT** belong to any department. Show their **last name** only. Use LEFT JOIN and then filter for NULL values.

```
Expected columns: last_name | Tables: employees LEFT JOIN departments |  
Extra: WHERE ... IS NULL
```

Your Answer:

Task 9. Count Per Department Including Zero

Difficulty: Medium

Write a query to show each **department name** and the **number of employees** in it. Use LEFT JOIN so that departments with **zero employees** also appear. Use COUNT and GROUP BY.

```
Expected columns: department_name, COUNT(*) | Tables: departments LEFT JOIN  
employees | Extra: GROUP BY, COUNT
```

Your Answer:

Task 10. Three-Table LEFT JOIN

Difficulty: Medium

Write a query to show the **department name** and the **city** where it is located. Use LEFT JOIN so that departments without a location also appear. Join **departments** with **locations**.

```
Expected columns: department_name, city | Tables: departments LEFT JOIN locations
```

Your Answer:

Task 11. Employee and Manager Name

Difficulty: Medium

Write a query to show the **employee last name** and their **manager's last name**. The manager is also in the employees table. The manager_id in employees points to another employee's employee_id. Use LEFT JOIN so that employees without a manager also appear.

```
Expected columns: employee last_name, manager last_name | Table: employees (self LEFT JOIN on manager_id = employee_id)
```

Your Answer:

Task 12. Average Salary by Department Including Empty

Difficulty: Hard

Write a query to show each **department name** and the **average salary** of its employees. Use LEFT JOIN to include departments with no employees. Use GROUP BY and ORDER BY the average salary from highest to lowest.

```
Expected columns: department_name, AVG(salary) | Tables: departments LEFT JOIN employees | Extra: GROUP BY, AVG, ORDER BY
```

Your Answer:

Part 3: RIGHT (OUTER) JOIN

RIGHT JOIN returns ALL rows from the right table. If there is no match in the left table, the result shows **NULL** for the left table columns. Tip: You can always rewrite a RIGHT JOIN as a LEFT JOIN by swapping the table order.

Task 13. Basic RIGHT JOIN

Difficulty: Easy

Write a query using RIGHT JOIN to show the **employee last name** and **department name**. ALL departments must appear, even those with no employees. Put **employees** on the LEFT and **departments** on the RIGHT.

```
Expected columns: last_name, department_name | Tables: employees RIGHT JOIN departments
```

Your Answer:

Task 14. Rewrite RIGHT JOIN as LEFT JOIN

Difficulty: Easy

Take the same query from Task 13 and rewrite it using only LEFT JOIN (no RIGHT JOIN). The result must be exactly the same. Show **department name** and **employee last name**.

```
Expected columns: department_name, last_name | Tables: departments LEFT JOIN employees (swapped order)
```

Your Answer:

Task 15. RIGHT JOIN with WHERE

Difficulty: Medium

Write a query to show **department name** and **employee last name**. Use RIGHT JOIN. Show only departments where the **location_id** is **1700**.

```
Expected columns: department_name, last_name | Tables: employees RIGHT JOIN  
departments | Extra: WHERE location_id = 1700
```

Your Answer:

Task 16. Employee Count with RIGHT JOIN

Difficulty: Hard

Write a query to show each **department name** and the **number of employees**. Use RIGHT JOIN so that all departments appear. Also add a WHERE clause to show only departments with **more than 5 employees**.

```
Expected columns: department_name, COUNT(*) | Tables: employees RIGHT JOIN  
departments | Extra: GROUP BY, HAVING COUNT > 5
```

Your Answer:

Part 4: FULL OUTER JOIN

FULL OUTER JOIN returns ALL rows from BOTH tables. When there is a match, rows are combined. When there is no match, the missing side shows NULL. You get everything from both tables.

Task 17. Basic FULL OUTER JOIN

Difficulty: Easy

Write a query to show **employee last name** and **department name**. Use FULL OUTER JOIN so that ALL employees and ALL departments appear. Employees without a department will show NULL for department_name. Departments without employees will show NULL for last_name.

```
Expected columns: last_name, department_name | Tables: employees FULL OUTER JOIN departments
```

Your Answer:

Task 18. FULL JOIN - Find Unmatched Rows

Difficulty: Medium

Write a query using FULL OUTER JOIN to find rows that do NOT match. Show the **employee last name** and **department name**. Include only rows where either the employee has no department OR the department has no employees.

```
Expected columns: last_name, department_name | Tables: employees FULL OUTER JOIN departments | Extra: WHERE ... IS NULL OR ... IS NULL
```

Your Answer:

Task 19. Five-Table JOIN (Employee Location Chain)

Difficulty: Hard

Write a query to show: **employee last name**, **department name**, **city**, **country name**, and **region name**. Join 5 tables: **employees**, **departments**, **locations**, **countries**, and **regions**. Use INNER JOIN for all steps. Order by **region name** and then **country name**.

```
Expected columns: last_name, department_name, city, country_name, region_name | Tables: employees, departments, locations, countries, regions
```


| Extra: ORDER BY

Your Answer:

Task 20. Six-Table JOIN (Full HR Report)

Difficulty: Hard

Write a query to create a complete HR report. Show: **employee last name**, **job title**, **department name**, **city**, **country name**, and **region name**. Join all 6 main HR tables: **employees**, **jobs**, **departments**, **locations**, **countries**, and **regions**. Use LEFT JOIN for all steps to keep all employees, even if some data is missing. Order by **department name**, then by **last name**.

Expected columns: last_name, job_title, department_name, city, country_name, region_name | Tables: employees, jobs, departments, locations, countries, regions | Extra: LEFT JOIN, ORDER BY

Your Answer:

Score Summary

Part	JOIN Type	Tasks	Difficulty
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1	INNER JOIN	1 - 6	Easy to Medium
2	LEFT JOIN	7 - 12	Easy to Hard
3	RIGHT JOIN	13 - 16	Easy to Hard
4	FULL OUTER JOIN	17 - 20	Easy to Hard
	Total: 20 Tasks		